

OFFICE OF THE MEDICAL EXAMINER

AUTOPSY REPORT

Case Number:	ME-2024-003-NAT
Decedent Name:	CONFIDENTIAL
Age/Sex/Race:	58-year-old White Male
Date/Time of Death:	November 28, 2024, 11:47 hours (pronounced)
Date/Time of Collapse:	November 28, 2024, 10:32 hours
Location of Collapse:	Thompson Manufacturing, 3200 Industrial Parkway
Date/Time of Autopsy:	November 29, 2024, 08:30 hours
Place of Autopsy:	County Medical Examiner Facility
Prosector:	Dr. Elizabeth Patterson, MD
Cause of Death:	Acute Myocardial Infarction
Manner of Death:	NATURAL

CIRCUMSTANCES

This 58-year-old male collapsed suddenly at his workplace while walking toward the break room. Witnesses observed him grasp his chest and complain of chest pain immediately before losing consciousness. Bystander CPR was initiated promptly, and emergency medical services responded. Despite aggressive resuscitation efforts at the scene, during transport, and in the emergency department, the patient never achieved return of spontaneous circulation and was pronounced dead at St. Mary's Hospital at 11:47 hours. This autopsy is performed to determine the cause and manner of death.

EXTERNAL EXAMINATION

General Description

The body is that of a well-developed, overweight adult White male appearing consistent with the stated age of 58 years. The body measures 175 centimeters (5 feet 9 inches) in length and weighs 98 kilograms (216 pounds), yielding a body mass index of 32, consistent with class I obesity. The body is clothed in work attire including jeans, work boots, and a button-down shirt. Rigor mortis is fully developed. Livor mortis is present and fixed in posterior dependent areas.

Evidence of Medical Intervention

Evidence of recent aggressive medical resuscitation is present and includes: endotracheal tube in place within the trachea, intravenous catheter sites in both antecubital fossae with surrounding ecchymosis, ECG electrode pads on the chest (removed for examination), automated external defibrillator pad marks on the chest, and scattered petechiae on the face and conjunctivae consistent with increased venous pressure during resuscitation. Minor amounts of dried blood are present around the mouth and nose, consistent with resuscitation efforts.

Head and Neck

The head is normocephalic and atraumatic. The scalp is intact without lacerations, contusions, or abrasions. The face shows no traumatic injuries. The conjunctivae demonstrate bilateral petechiae consistent with resuscitation efforts. The neck is symmetrical without external trauma. No ligature marks or signs of manual compression are present.

Trunk and Extremities

The chest is symmetrical with increased anterior-posterior diameter consistent with body habitus. No rib fractures are palpable (rib fractures from CPR are common but not present in this case). The abdomen is protuberant consistent with obesity. All four extremities are intact without acute traumatic injuries. No defensive wounds, injection sites, or other signs of trauma are identified. Minor healing abrasions on both knees are noted, consistent with normal work-related activities.

INTERNAL EXAMINATION

Body Cavities

The body cavities are entered through standard Y-shaped incision. Subcutaneous tissue demonstrates increased adiposity consistent with obesity. The pleural cavities contain small amounts of clear serous fluid bilaterally (approximately 30 mL each). The pericardial sac contains approximately 25 mL of clear serous fluid. The peritoneal cavity is unremarkable. Organs are in normal anatomic position.

Cardiovascular System - Detailed Examination

Heart: The heart weighs 520 grams, significantly elevated above normal range (normal adult male 270-360 grams), indicating cardiac hypertrophy. The epicardial surface shows moderate adiposity but is otherwise smooth and without adhesions. The pericardium is intact.

Coronary Arteries - Critical Findings: The coronary arteries are dissected and examined carefully. The arteries demonstrate severe atherosclerotic disease:

LEFT ANTERIOR DESCENDING (LAD) ARTERY: Serial sections reveal severe atherosclerotic disease throughout the vessel. At the mid-vessel level, approximately 4 cm from the origin, there is critical stenosis with approximately 90% luminal narrowing caused by a large, complex atherosclerotic plaque. Most significantly, an acute thrombus is present at this location, adherent to the plaque surface and completely occluding the vessel lumen. The thrombus appears fresh (red-brown color, soft consistency) and measures approximately 1.2 cm in length. The underlying plaque shows evidence of rupture with disruption of the fibrous cap.

LEFT CIRCUMFLEX ARTERY: Demonstrates moderate atherosclerotic disease with approximately 60% stenosis in the proximal segment. No acute thrombus is present.

RIGHT CORONARY ARTERY: Shows severe atherosclerotic disease with 75-80% stenosis in the proximal segment. Multiple calcified plaques are present. No acute thrombus identified.

Myocardium: The left ventricular wall measures 2.0 cm in thickness (normal 0.9-1.4 cm), confirming left ventricular hypertrophy. The right ventricular wall measures 0.4 cm (normal 0.2-0.4 cm). Examination of the left ventricular myocardium reveals an area of pallor and slight firmness in the anterior wall and interventricular septum, measuring approximately 4 x 3 cm. This region corresponds to the territory supplied by the LAD artery. The affected tissue shows early changes of acute infarction including slight discoloration and loss of normal texture. The changes are subtle, consistent with an infarction occurring within hours of examination.

Serial sections through the remainder of the left ventricular myocardium reveal several small, healed scars (old infarcts) measuring 0.3-0.5 cm, visible as gray-white areas of fibrosis. These represent previous ischemic events that were apparently clinically silent or unrecognized. The right ventricular myocardium appears normal.

Cardiac Chambers and Valves: All four cardiac chambers are dilated, particularly the left ventricle, consistent with chronic cardiac stress. The endocardium is smooth. All cardiac valves (aortic, mitral, tricuspid, pulmonic) are examined and found to be structurally normal without significant pathology. The valves are thin, pliable, and competent. No vegetations, calcifications, or abnormalities are present.

Aorta and Great Vessels: The aorta demonstrates moderate atherosclerotic disease with scattered plaques and mild luminal narrowing, appropriate for age. No aneurysm, dissection, or acute pathology is identified. The superior and inferior vena cavae are patent and unremarkable.

Respiratory System

The larynx, trachea, and bronchi are patent and lined with normal mucosa. The endotracheal tube is positioned correctly. The right lung weighs 680 grams and the left lung weighs 660 grams (combined weight elevated above normal). Both lungs demonstrate moderate pulmonary edema with pink-tan parenchyma and frothy fluid exuding from cut surfaces. The congestion and edema are secondary to acute cardiac failure. No pneumonia, masses, emboli, or primary pulmonary disease is identified. The pleural surfaces are smooth without adhesions.

Hepatobiliary System

The liver weighs 1,980 grams (normal 1,400-1,800 grams), mildly enlarged. The capsular surface is smooth. The parenchyma demonstrates moderate passive congestion with prominent central veins,

consistent with cardiac failure. The texture is normal. No masses, cirrhosis, or significant pathology is present. The gallbladder contains bile and no stones. The biliary tree is patent.

Gastrointestinal System

The esophagus, stomach, small intestine, and large intestine are unremarkable without masses, ulcers, or significant pathology. The stomach contains approximately 150 mL of partially digested food material. The pancreas weighs 95 grams and appears normal without masses or inflammation.

Genitourinary System

The right kidney weighs 155 grams and left kidney weighs 165 grams (combined normal 240-340 grams). Both kidneys have smooth capsular surfaces. On sectioning, the kidneys show mild nephrosclerosis with decreased corticomedullary distinction, consistent with hypertension and age-related changes. No masses, cysts, or acute pathology is identified. The renal pelvis and ureters are unremarkable. The urinary bladder is unremarkable. The prostate shows benign enlargement appropriate for age.

Other Organ Systems

The spleen weighs 180 grams and is unremarkable. The thyroid, adrenal glands, and pituitary gland appear normal. Lymph nodes throughout the body are not enlarged. No significant pathology is identified in any other organ system.

Central Nervous System

The scalp and skull are intact without trauma. The dura mater is intact and unremarkable. No epidural or subdural hemorrhages are present. The brain weighs 1,440 grams (normal 1,300-1,400 grams). The brain appears grossly normal without mass effect, herniation, or focal lesions. The leptomeninges are clear. Blood vessels at the base of the brain are patent without aneurysm or significant atherosclerosis. Coronal sections reveal normal gray-white matter differentiation without hemorrhage, infarction, or masses. The cerebellum and brainstem are unremarkable.

Musculoskeletal System

The ribs are intact without fractures despite CPR. The spine shows moderate degenerative changes appropriate for age without acute injury. Long bones are intact. Skeletal muscle is normal.

MICROSCOPIC EXAMINATION

Representative sections of organs are processed in routine fashion with hematoxylin and eosin staining. Microscopic examination reveals:

Heart - Acute Infarction Zone: Sections from the pale area of left ventricular myocardium demonstrate classic histological features of early acute myocardial infarction. Myocytes show coagulative necrosis with loss of nuclei, hypereosinophilia of cytoplasm, and loss of cross-striations. Interstitial edema is prominent. Early inflammatory response is evidenced by neutrophilic infiltration beginning at the margins of the necrotic zone. These findings confirm acute myocardial infarction.

occurring within approximately 4-12 hours before death.

Heart - Non-infarcted Myocardium: Sections from non-infarcted regions show left ventricular hypertrophy with increased myocyte diameter. Interstitial fibrosis is present, representing chronic changes from hypertension and previous ischemic injury. Small foci of replacement fibrosis (healed scars) are identified, confirming that the heart had experienced previous ischemic events.

Coronary Arteries: Sections through the coronary arteries confirm severe atherosclerotic disease. The plaques are complex with large lipid cores, fibrous caps of varying thickness, areas of calcification, and regions of inflammation. The thrombus in the LAD shows layering and fibrin deposition consistent with acute formation. The section demonstrates plaque rupture with disruption of the fibrous cap, exposing thrombogenic lipid core to the vessel lumen and triggering thrombus formation.

Lungs: Sections demonstrate moderate pulmonary edema with proteinaceous fluid in alveolar spaces and congestion of capillaries. No pneumonia, fibrosis, or neoplasm is present.

Liver: Sections show passive congestion with dilated central veins and compressed hepatocytes in central zones, consistent with cardiac failure. No significant primary hepatic pathology is present.

Kidneys: Sections demonstrate nephrosclerosis with thickening of arteriolar walls, glomerulosclerosis, and interstitial fibrosis, consistent with hypertensive changes. No acute tubular necrosis or glomerulonephritis is present.

TOXICOLOGY

Blood and urine specimens submitted for comprehensive toxicological analysis. Blood alcohol: negative. Comprehensive drug screen: negative for all tested substances. See separate toxicology report for complete findings.

OPINION

This 58-year-old male died of **acute myocardial infarction due to acute thrombotic occlusion of the left anterior descending coronary artery due to severe atherosclerotic coronary artery disease**. The manner of death is **NATURAL**.

The autopsy findings demonstrate severe three-vessel coronary artery disease with critical stenosis and acute thrombotic occlusion of the LAD artery. The thrombus formation resulted from rupture of an atherosclerotic plaque, exposing thrombogenic material that triggered acute vessel occlusion. The sudden loss of blood supply to a large territory of the left ventricle resulted in acute myocardial infarction, confirmed by both gross and microscopic examination.

The left ventricular hypertrophy indicates chronic cardiac stress from hypertension. The presence of old healed myocardial scars indicates the heart had experienced previous ischemic events. The constellation of findings represents long-standing coronary artery disease that culminated in a fatal acute event.

The investigation excluded trauma, poisoning, and other non-natural causes. The death resulted from natural disease processes (atherosclerotic cardiovascular disease) without external causes. The manner of death is appropriately classified as natural.

Other Significant Conditions: Hypertensive cardiovascular disease with left ventricular hypertrophy, obesity (BMI 32), nephrosclerosis.

Elizabeth Patterson, MD

Chief Medical Examiner

Board Certified in Anatomic and Forensic Pathology

Date: November 30, 2024